Möbius-Admissible Completions, Discrepancy, and RH-Critical Excursions: A Literature and Research Map

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Abstract

This document is a research map for a deterministic extremal program around the Möbius summatory function. The starting space is the full ternary cube $\{-1,0,1\}^N$. Successive layers impose rigorous Möbius-specific information—squarefree support, multiplicative compatibility, partial prime information, residual closure, identities, local pattern constraints, and other arithmetic consequences—and the surviving completions are tested by the largest partial sum still admissible. The purpose of the present document is not to claim novelty for the ingredients individually. Most ingredients belong to well-developed literatures on multiplicative functions, discrepancy, random multiplicative functions, sign patterns, and the Mertens function. Its role is to identify precisely which parts are known, which results can be converted into deterministic constraints, and where a genuinely different extremal-completion problem may remain.

1 The organizing problem

Let

$$\Omega_N^{(0)} = \{-1,0,1\}^N, \qquad |\Omega_N^{(0)}| = 3^N,$$

and let $\Omega_N^{(j)} \subseteq \Omega_N^{(0)}$ be the set of words surviving a specified collection $\mathcal{I}_j$ of rigorous arithmetic constraints. For $\omega = (\omega_1, \dots, \omega_N)$ put

$$S_\omega(n) = \sum_{k \le n} \omega_k, \qquad B_N^{(j)} = \max_{\omega \in \Omega_N^{(j)}} |S_\omega(N)|.$$

The corresponding effective extremal exponent is

$$E_N^{(j)} = \frac{\log B_N^{(j)}}{\log N} - \frac{1}{2}$$

when $B_N^{(j)} > 0$. The central deterministic question is not whether a large excursion is improbable, but whether any arithmetically admissible completion producing it remains.

A useful filtration has the schematic form

$$\Omega_N^{(0)} \supseteq \Omega_N^{(\mathrm{sqf})} \supseteq \Omega_N^{(\mathrm{mult})} \supseteq \Omega_N^{(P_r)} \supseteq \Omega_N^{(\mathrm{id})} \supseteq \cdots,$$

where the precise information content of every layer must be stated independently of the actual unknown values one is attempting to bound. Imposing the complete definition of Möbius together with complete factorization reduces the space to the unique true Möbius word and is therefore an endpoint check, not a nontrivial layer of the program.

2 RHC, RHFc, and RHFl

For a finite word, define its endpoint excess exponent by

$$e_N(\omega) = \frac{\log |S_\omega(N)|}{\log N} - \frac{1}{2},$$

when $S_\omega(N) \neq 0$.

RHC. An RHC (RH-critical) event is a finite excursion above the square-root scale,

$$|S_\omega(N)| > \sqrt{N}.$$

Such an event is not incompatible with RH. Factors of size $N^{o(1)}$, including powers of iterated logarithms, may produce arbitrarily large ratios to $\sqrt{N}$ while retaining exponent $1/2 + o(1)$.

RHFc. For fixed $\delta > 0$, an RHFc$_\delta$ event is the finite candidate condition

$$|S_\omega(N)| > N^{1/2+\delta}.$$

A single such event at finite N does not falsify RH. It becomes asymptotically incompatible with RH only if a fixed positive excess survives at unbounded scales in a manner forcing

$$\limsup_{N\to\infty} \frac{\log |M(N)|}{\log N} > \frac{1}{2}.$$

RHFl. An RHFl event is genuinely local: it violates an explicit finite-x consequence of RH. If a theorem gives

$$\mathrm{RH} \Longrightarrow |M(x)| \leq H_{\mathrm{RH}}(x)$$

in an explicit range, then $|S_\omega(N)| > H_{\mathrm{RH}}(N)$ is locally incompatible with RH in that range. Existing explicit RH-conditional estimates have thresholds far beyond the initial computational regime, so RHFl and RHFc must not be conflated.

3 Classical Mertens-function background

RH is equivalent to

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \qquad \text{for every } \varepsilon > 0.$$

This is an asymptotic statement with an ε-dependent constant; therefore no unit-constant inequality $|M(N)| \leq N^{1/2+\varepsilon}$ at one finite N is an RH consequence. Ng studies the conditional distribution and extreme-value behavior of $M(x)$ under RH together with additional hypotheses, including a limiting distribution for the normalized summatory function and a heuristic extreme scale involving $\sqrt{x}(\log\log\log x)^{5/4}$ [23]. Explicit conditional estimates, such as those of Simonič, are useful for defining a genuine RHFl boundary but only in their stated ranges [24].

4 Discrepancy and adversarial multiplicativity

The discrepancy literature is essential because it is deterministic and adversarial rather than probabilistic. Tao proved the Erdős discrepancy problem: every $\{\pm 1\}$ sequence has unbounded discrepancy on homogeneous arithmetic progressions [1]. For a completely multiplicative $\{\pm 1\}$-valued sequence, discrepancy on a homogeneous progression factors through the ordinary partial sum, so the multiplicative specialization asks whether ordinary partial sums can remain bounded. Earlier and related work studies multiplicative functions with bounded or forced-unbounded partial sums [2, 3].

The conceptual lesson for the present program is sharp: a severe reduction in the number of admissible multiplicative configurations need not imply a comparable reduction in the extremal partial sum. Discrepancy results force excursions under structural constraints; the desired Möbius-side counterpart would be an upper-bound or contraction theorem showing that sufficiently strong Möbius-specific constraints forbid excursions of fixed power above $N^{1/2}$.

5 Structure of multiplicative functions with small sums

Koukoulopoulos and Soundararajan characterize strong structural consequences of unusually small partial sums of multiplicative functions [4]. This literature is close to the contrapositive viewpoint of the present program: instead of asking what structure follows from small sums, one asks how much a prescribed structure restricts the largest possible sum. Any claimed novelty must therefore be phrased at the level of *partial-information completion* and the quantitative contraction of its extremal envelope, not merely as an extremal problem over multiplicative functions.

6 Möbius-like multiplicative functions on squarefree and k-free support

Aymone gives explicit multiplicative functions supported on the squarefree integers, with prime values $f(p) = \pm 1$, whose summatory functions satisfy $M_f(x) = o(\sqrt{x})$; under RH the construction can go beyond square-root cancellation [9]. Aymone, Bueno, and Medeiros extend this to multiplicative functions supported on the k-free integers, constructing examples with partial sums $o(x^{1/k})$ and stronger conditional exponents under GRH [10]. Aymone also studies the Erdős discrepancy problem on squarefree and cubefree supports, proving unbounded partial sums for natural multiplicative sequences built from these supports [11].

These papers are especially important for the present program because they show that the layer "squarefree support + multiplicativity + free prime signs" already belongs to an established extremal literature. Therefore a new contribution cannot consist merely of optimizing over prime signs on squarefree support. Any distinct object must use genuinely Möbius-specific information closure or quantify the marginal effect of information that these model classes do not encode.

7 Random multiplicative functions as a scale reference

Wintner's Rademacher random multiplicative model is a classical probabilistic model for Möbius. Modern work gives a detailed picture of its large fluctuations. Harper proves almost sure arbitrarily large fluctuations at scale

$$\sqrt{x}(\log\log x)^{1/4+o(1)},$$

showing in particular why crossing $\sqrt{x}$ is not itself an RH-false phenomenon [7]. Harper's high-moment work gives large-deviation information for Rademacher and Steinhaus random multiplicative sums [6]. Recent upper-bound work determines the matching logarithmic scale in these models [8].

These results are not evidence that RH is "probable." Their role here is diagnostic: they identify a natural multiplicative fluctuation scale of the form $x^{1/2+o(1)}$ and therefore provide a reference against which deterministic RHC and RHFc envelopes can be interpreted.

8 Biased random multiplicative functions and RH-equivalent model statements

Aymone and Sidoravicius study random multiplicative functions supported on the squarefree integers whose prime values are biased toward the classical Möbius function [12]. Their results include necessary and sufficient conditions for power-saving partial sums in biased models and an RH-equivalent statement formulated through a family of weakly biased random multiplicative functions. This is a particularly close antecedent to any attempt to interpret progressive Möbius information probabilistically.

For the present deterministic program the distinction is decisive: conditioning or biasing a probability law on prime values does not eliminate every exceptional completion. Nevertheless, this literature supplies both warning and calibration: a filtration whose only new ingredient is a growing bias toward $\mu(p) = -1$ is unlikely to constitute a distinct research problem.

9 Local sign patterns and realizability

Matomäki, Radziwiłł, and Tao prove positive-density occurrence results for local sign patterns of Liouville and for the nine possible pairs $(\mu(n), \mu(n+1))$ [5]. This addresses local realizability of Möbius patterns, whereas the present completion problem concerns global prefixes $(1, \dots, N)$ subject to simultaneous arithmetic constraints. Nevertheless, sign-pattern theorems can become genuine constraints when a filtration contains translated windows or local-pattern requirements; they must not be reintroduced as if they were new combinatorial observations.

10 Partial multiplicativity and prescribed prime information

Vandehey explicitly studies weakened multiplicativity in which multiplicative relations are imposed only for primes in a finite prescribed set [3]. This is a direct antecedent of any layer parameterized merely by a finite set of known primes. Consequently, "reveal the first r primes and optimize" is not by itself a sufficiently distinct research object.

The potentially different object is instead the *closure* of partial Möbius information: after revealing a specified information set, include every value, zero, sign relation, residual primality conclusion, and multiplicative identity that follows logically from it, and optimize over all completions consistent with that closure. The mathematical object is therefore a constraint-closure operator rather than a list of fixed prime signs.

11 Pretentiousness and inverse information from partial sums

The pretentious approach of Granville and Soundararajan organizes multiplicative functions through a quantitative distance measuring how closely one multiplicative function imitates another [13]. Koukoulopoulos–Soundararajan and subsequent work of Sachpazis develop inverse principles in which unusually small partial sums force strong average structure on prime values [4, 14]. This literature is relevant to the research map in two directions. First, it shows that partial-sum behavior itself carries information about the underlying prime data. Second, it provides candidate analytic constraints that could eventually replace brute-force propagation by structural inequalities.

A future completion theorem should therefore be compared not only with discrepancy theory but also with these inverse theorems: if a proposed "new" constraint is already a consequence of pretentious distance or an inverse theorem for small sums, the literature result should be used directly.

12 Computational Mertens benchmarks

Direct computation of the true Mertens function is not the objective of the completion program, but it supplies indispensable controls. Hurst computed all values through 10^{16} and improved numerical bounds associated with the disproved Mertens conjecture [21]. A 2026 implementation computes isolated values through 10^{25}, including $M(10^{24})$ and $M(10^{25})$, while retaining the classical $O(x^{2/3+\varepsilon})$ isolated-value framework [22]. These computations make clear that finite verification of the true word can be pushed vastly farther than exhaustive enumeration of completion spaces. Consequently, computational novelty must come from solving the adversarial completion problem, not from tabulating $M(N)$ itself.

13 Chudakov-type classification and bounded partial sums

A further novelty boundary comes from classification theorems for multiplicative functions with bounded partial sums. Klurman proves Chudakov's conjecture in a finite-range setting: under the appropriate hypotheses, a completely multiplicative function with bounded partial sums is forced into character-like structure [15]. Klurman, Mangerel, Pohoata, and Teräväinen subsequently develop sieve-theoretic tools for multiplicative functions whose partial sums stay close to their mean and obtain, among other consequences, strong rigidity statements for bounded partial sums [16]. Aymone gives additional classifications and constructions for complex-valued multiplicative functions with bounded partial sums [17].

These results occupy a large part of the naive extremal question "which multiplicative assignments have bounded sums?" The present program must therefore retain its finite-information aspect: the variable object is not a completed multiplicative function selected from a broad class, but the adversarial envelope of all completions consistent with a declared, incomplete Möbius information state.

14 Finite prime modifications and stability of partial sums

Aymone studies completely multiplicative functions obtained by modifying a primitive Dirichlet character at a finite set of primes, proving Ω-bounds whose scale depends explicitly on the number of modified primes [18]. This is very close to any experiment in which finitely many prime values

are altered while all remaining prime data are frozen. It shows that the location and number of modified prime generators can leave a quantitative signature in the partial sums.

Consequently, a layer obtained merely by "freeing" or "fixing" finitely many prime signs is not enough to define a new object. The potentially distinct ingredient is again the closure map: partial information must generate all deterministic Möbius consequences before the adversarial maximum is taken.

15 Further Möbius-like extremal constructions

Liu studies classes of multiplicative functions resembling Möbius and proves complementary O- and Ω-results for their summatory functions [19]. Together with the squarefree and k-free constructions already reviewed, this reinforces the point that extremizing partial sums over Möbius-like multiplicative families is itself established territory. The completion program should use these papers as test families: any proposed envelope theorem should be checked against known constructions that deliberately produce unusually small or unusually large sums.

16 Exponential-sum rigidity with multiplicative coefficients

Recent work of Bazin studies bounded exponential sums with multiplicative coefficients and develops structural consequences from boundedness, extending ideas related to Chudakov-type rigidity to more general oscillatory observables [20]. Although the observable differs from the ordinary Mertens sum, this literature is relevant methodologically: it supplies additional examples where a global bound forces pretentious or character-like structure. Such inverse mechanisms may eventually be convertible into deterministic pruning rules for a completion algorithm.

17 Targeted search for completion, query, and certificate formulations

A dedicated search was carried out for formulations under terms such as *partial specification of multiplicative functions*, *extension/completion of multiplicative functions*, *reconstruction of Möbius from partial values*, *query complexity of multiplicative functions*, and *certificate complexity for Möbius*. The closest primary-source results located were partial multiplicativity on prescribed prime sets, inverse theorems from small partial sums, rigidity/classification results for bounded sums, finite modifications of characters, and probabilistic conditioning on prime values. No primary-source paper was located that defines the finite adversarial completion functional

$$\mathfrak{B}_N(\mathcal{I}) = \max_{\omega \in \mathcal{C}_N(\mathcal{I})} \left| \sum_{n \le N} \omega_n \right|$$

for a progressively closed body of Möbius-specific information and then studies its contraction as an information-complexity problem. This is a negative search result, not a proof of novelty, and should be updated as the bibliography grows.

18 Novelty-threat matrix

The following table is intended as a guardrail against rediscovery.

Candidate idea	Closest established literature	Status	What would still be different
Free signs on squarefree support	Aymone; Aymone–Bueno–Medeiros; Liu	Occupied	Möbius-specific information closure
Finite set of prescribed prime values	Vandehey; modified-character literature	Occupied	Optimize after full logical closure, not after raw prescriptions
Bounded/unbounded multiplicative partial sums	Tao; Klurman; KMPT; Aymone	Occupied	RH-scale upper envelope under partial information
Probabilistic progressive prime information	Aymone–Sidoravicius; random multiplicative literature	Occupied	Deterministic elimination of every surviving completion
Inverse structure from small sums	Koukoulopoulos–Soundararajan; Sachpazis; Bazin	Occupied	Forward contraction from declared finite information
Count of surviving words	Elementary combinatorics / CSP viewpoint	Not sufficient	Coupled with arithmetic closure and extremal envelope
Adversarial Möbius completion envelope	No exact antecedent located in targeted search	Open search status	Formalize closure, information cost, and N-scaling
Minimal information killing every RHFc$_\delta$ completion	No exact antecedent located in targeted search	Open search status	Query/certificate complexity of RH-critical exclusion

19 The completion operator

Let $\mathcal{I}$ be a finite body of arithmetic information available at scale N. Define

$$\mathcal{C}_N(\mathcal{I}) = \{\omega \in \{-1,0,1\}^N : \omega \text{ satisfies every consequence of } \mathcal{I} \text{ included in the model}\}.$$

The fundamental extremal functional is

$$\mathfrak{B}_N(\mathcal{I}) = \max_{\omega \in \mathcal{C}_N(\mathcal{I})} \left| \sum_{n \le N} \omega_n \right|.$$

For nested information sets $\mathcal{I} \subseteq \mathcal{J}$,

$$\mathcal{C}_N(\mathcal{J}) \subseteq \mathcal{C}_N(\mathcal{I}), \qquad \mathfrak{B}_N(\mathcal{J}) \le \mathfrak{B}_N(\mathcal{I}).$$

The contraction produced by adding information is

$$\Delta_{\mathcal{I}\to\mathcal{J}}(N) = \mathfrak{B}_N(\mathcal{I}) - \mathfrak{B}_N(\mathcal{J}),$$

or, on the exponent scale,

$$\Delta E_{\mathcal{I}\to\mathcal{J}}(N) = \frac{\log \mathfrak{B}_N(\mathcal{I}) - \log \mathfrak{B}_N(\mathcal{J})}{\log N}.$$

This distinguishes information that merely reduces the number of completions from information that actually contracts the dangerous extremal envelope.

20 Constraint families to be tested

The following layers should be treated as separate mathematical experiments.

1. Full ternary space $\{-1, 0, 1\}^N$.
2. Möbius support information: $\mu(1) = 1$, squarefree/non-squarefree constraints.
3. Multiplicative compatibility with prime signs still free.
4. Finite-prime partial multiplicativity, explicitly compared with the Vandehey-type antecedent.
5. Sieve information available below a moving threshold, especially $p \le \sqrt{N}$.
6. Residual closure by the first r primes: factor extraction plus every forced residual-primality conclusion licensed by the chosen closure rule.
7. Möbius identities that couple many coordinates without revealing them individually.
8. Prefix or endpoint information, when it is part of the experiment, with its information cost recorded separately.
9. Proven local pattern/correlation constraints when they can be encoded deterministically.
10. The full Möbius word only as a terminal control, never as evidence that an intermediate filtration contracts.

21 Quantities to record at every layer

For every (N, j) the computational record should contain at least

$$|\Omega_N^{(j)}|, \qquad B_N^{(j)}, \qquad E_N^{(j)},$$

and the exact counts

$$\#\{\omega \in \Omega_N^{(j)} : |S_\omega(N)| > \sqrt{N}\},$$
$$\#\{\omega \in \Omega_N^{(j)} : |S_\omega(N)| > N^{1/2+\delta}\}$$

for a declared grid of fixed $\delta > 0$. Counts are descriptive; the logical RH question depends on emptiness of the relevant admissible set. The denominator 3^N should always be retained as the reference size of the original unconstrained space, while conditional proportions inside each surviving layer may be recorded separately.

For pathwise experiments, endpoint constraints must be supplemented by persistence statistics, for example the number of increasing scales at which a fixed δ is exceeded. RHF itself is asymptotic and cannot be certified by a single finite endpoint.

22 Literature-to-constraint map

Literature	Object / mode	Direct use here	Main limitation
Tao [1]	Discrepancy of ± 1 sequences; deterministic	Forced excursions; adversarial benchmark	Not a square-root/RH-scale upper bound
Vandehey [3]	Partial multiplicativity on a finite prime set; deterministic	Antecedent for partial-prime layers	Does not build Möbius-specific closure
Koukoulopoulos–Soundararajan [4]	Multiplicative functions with small partial sums; structural	Benchmark for extremal sums and inverse constraints	Reverse direction from completion contraction
Matomäki–Radziwiłł–Tao [5]	Local Liouville/Möbius sign patterns; deterministic	Realizability and local constraints	Translated local windows, not global prefix completions
Harper [7, 6]	Random multiplicative partial sums; probabilistic	Natural $x^{1/2+o(1)}$ scale and large deviations	Cannot eliminate exceptional deterministic completions
Durkan–Pearce-Crump [8]	Sharp a.s. upper scale in random models	Reference scale for RHC	Random model, not Möbius
Aymone [9]	Squarefree-supported Möbius-like functions; constructive	Shows free prime-sign squarefree layer is established territory	No progressive Möbius-specific closure
Aymone–Bueno–Medeiros [10]	k-free supported multiplicative functions; constructive	Extremal cancellation benchmark on restricted support	Different information model
Aymone–Sidoravicius [12]	Biased random Möbius-like functions	Warning for progressive prime bias; RH-equivalent model results	Exceptional deterministic completions remain
Granville–Soundararajan [13]	Pretentious distance; analytic	Structural metric for prime information	Not a finite completion envelope
Sachpazis [14]	Inverse structure from small partial sums	Candidate source of global constraints	Reverse/inverse direction
Klurman [15]	Chudakov-type rigidity; completely multiplicative functions	Classification benchmark for bounded sums	Completed functions, not partial-information envelopes
Klurman–Mangerel–Pohoata–Teräväinen [16]	Sieve/inverse structure for multiplicative partial sums	Rigidity and potential pruning inequalities	Does not optimize finite Möbius completions
Aymone [18]	Finite modifications of Dirichlet characters	Quantifies effects of finitely changed prime values	Remaining prime data are not an unknown completion state
Liu [19]	Möbius-like multiplicative functions; O/Ω bounds	Stress-test family for envelope claims	No progressive information filtration
Bazin [20]	Bounded exponential sums with multiplicative coefficients	Further inverse/rigidity mechanisms	Different observable and no Möbius completion functional
Hurst [21, 22]	Direct computation of $M(x)$	Ground truth and scaling benchmark	Computes the unique true word, not admissible completions
Ng [23]	Conditional distribution of $M(x)$	RH-compatible extreme scale	Uses RH plus additional hypotheses
Simonič [24]	Explicit conditional bounds for $M(x)$	Genuine RHFl boundary where applicable	Very large validity thresholds

23 Gap analysis

The following claims are *not* candidates for novelty by themselves: enumerating ternary words; imposing squarefree support; parameterizing multiplicative functions by prime values; conditioning or prescribing finitely many prime values; asking whether multiplicative partial sums are bounded; studying large deviations of random multiplicative sums; or proving that particular local Möbius sign patterns occur.

A potentially distinct problem remains if all of the following are essential:

1. the objects are finite Möbius-admissible *completions*, not a probability model;
2. information is represented by an explicit closure operator that propagates every permitted deterministic consequence;
3. the objective is the adversarial maximum $\mathfrak{B}_N(\mathcal{I})$ over all surviving completions;
4. the information content is varied systematically, allowing marginal contraction to be assigned to particular arithmetic constraints;
5. the target scale is the disappearance of every fixed-power excess $N^{1/2+\delta}$, not merely bounded discrepancy or small average behavior.

A literature claim of novelty should be made only after a dedicated search for extremal completion, partial specification, reconstruction/query complexity, and constraint-satisfaction formulations for multiplicative functions.

24 Research questions

The initial program is organized around the following questions.

1. Which known Möbius constraints reduce only $|\Omega_N^{(j)}|$, and which actually reduce $B_N^{(j)}$?
2. Are the contraction points of $B_N^{(j)}$ controlled by primes, prime squares, large prime factors, residual-primality transitions, or another arithmetic statistic?
3. Does a stable law emerge for $E_N^{(j)}$ as N grows for any nontrivial family of information sets?
4. Can one prove a power saving $B_N^{(j)} \ll N^{1-\eta}$ from a genuinely partial, generic information family?
5. Can any explicit growing family $\mathcal{I}(N)$ force
$$\mathfrak{B}_N(\mathcal{I}(N)) \le N^{1/2+o(1)}$$
without encoding the Möbius word itself?
6. What is the minimal information, under a declared information model, required to make the RHFc_δ set empty?
7. Which constraints are redundant for the extremal envelope even when they remove most admissible completions?

25 Computational protocol

Small-N exhaustive enumeration is a validation instrument, not the intended asymptotic method. Every implementation should reproduce exact counts from the 3^N reference space and then replace enumeration by state merging, constraint propagation, branch-and-bound, SAT/CSP methods, or analytic extremal bounds. Two completions that are indistinguishable with respect to all future constraints and the objective should be merged into one state. The central computational output is the extremal envelope, while exact multiplicities are retained as secondary information.

The first benchmark range $N \leq 25$ is useful because all layers can be audited exactly. Scaling beyond this range should occur only after the filtration has been checked for mathematical nonredundancy and the literature map has identified which constraints are already subsumed by known theorems.

26 Decision criterion for continuing the program

The program should be stress-tested before large-scale computation. If increasingly strong generic Möbius constraints leave

$$E_N^{(j)} \approx \frac{1}{2},$$

then the admissible maximum remains nearly linear and the known constraints provide little leverage toward RH. A stable contraction to an exponent strictly below 1 would already be a nontrivial extremal phenomenon. A systematic drift of the strongest genuinely partial-information envelope toward

$$E_N^{(j)} = 0$$

would justify a deeper analytic program aimed at proving the contraction law.

27 Working checklist

Before adding a new layer R:

1. identify the theorem or elementary arithmetic fact that licenses R;
2. search whether the same partial-information model has already been studied;
3. specify exactly what information R reveals and what it does not reveal;
4. compute the closure of R under the rules already admitted;
5. record the change in $|\Omega_N|$, B_N, E_N, RHC counts, and RHFc_δ counts;
6. distinguish a reduction in multiplicity from a reduction in the extremal envelope;
7. attempt an analytic explanation of every persistent contraction point.

28 Canonical finite benchmark for $1 \leq N \leq 25$

This section fixes a single reproducible finite benchmark. All counts are exact. The unrestricted ternary layer is computed by coefficient convolution of $(z^{-1} + 1 + z)^N$; the squarefree-support layer fixes $\mu(1) = 1$, fixes zero on nonsquarefree indices, and leaves the remaining squarefree signs free;

the multiplicative layer assigns free signs to the primes $p \le N$ and extends multiplicatively on squarefree integers; the last nonterminal layer additionally fixes only $\mu(2) = -1$. No true values $M(N)$ are used in these four layers.

The strict endpoint classes used here are

$$\mathrm{RHC}_N = \{\omega : |S_\omega(N)| > \sqrt{N}\}, \qquad \mathrm{RHFc}_{N,\delta} = \{\omega : |S_\omega(N)| > N^{1/2+\delta}\}.$$

The original reference population 3^N is retained in every table. A terminal control layer obtained by imposing the full prime rule $\mu(p) = -1$ for every prime $p \le N$ contains exactly one word and is therefore excluded from the nontrivial filtration.

28.1 Surviving completion counts

N	3^N	squarefree support	multiplicative	multiplicative $+\mu(2) = -1$
1	3	1	1	1
2	9	2	2	1
3	27	4	4	2
4	81	4	4	2
5	243	8	8	4
6	729	16	8	4
7	2,187	32	16	8
8	6,561	32	16	8
9	19,683	32	16	8
10	59,049	64	16	8
11	177,147	128	32	16
12	531,441	128	32	16
13	1,594,323	256	64	32
14	4,782,969	512	64	32
15	14,348,907	1,024	64	32
16	43,046,721	1,024	64	32
17	129,140,163	2,048	128	64
18	387,420,489	2,048	128	64
19	1,162,261,467	4,096	256	128
20	3,486,784,401	4,096	256	128
21	10,460,353,203	8,192	256	128
22	31,381,059,609	16,384	256	128
23	94,143,178,827	32,768	512	256
24	282,429,536,481	32,768	512	256
25	847,288,609,443	32,768	512	256

28.2 Extremal envelope and effective exponent

	ternary		squarefree		multiplicative		$+\mu(2) = -1$	
N	B_N	E_N	B_N	E_N	B_N	E_N	B_N	E_N
1	1	–	1	–	1	–	1	–
2	2	+0.50000	2	+0.50000	2	+0.50000	0	–
3	3	+0.50000	3	+0.50000	3	+0.50000	1	-0.50000
4	4	+0.50000	3	+0.29248	3	+0.29248	1	-0.50000
5	5	+0.50000	4	+0.36135	4	+0.36135	2	-0.06932
6	6	+0.50000	5	+0.39824	5	+0.39824	1	-0.50000
7	7	+0.50000	6	+0.42078	6	+0.42078	2	-0.14379
8	8	+0.50000	6	+0.36165	6	+0.36165	2	-0.16667
9	9	+0.50000	6	+0.31546	6	+0.31546	2	-0.18454
10	10	+0.50000	7	+0.34510	7	+0.34510	1	-0.50000
11	11	+0.50000	8	+0.36719	8	+0.36719	2	-0.21094
12	12	+0.50000	8	+0.33683	8	+0.33683	2	-0.22106

N	B_0	E_0	B_{sqf}	E_{sqf}	B_{mult}	E_{mult}	B_2	E_2
13	13	+0.50000	9	+0.35663	9	+0.35663	3	-0.07168
14	14	+0.50000	10	+0.37250	10	+0.37250	2	-0.23735
15	15	+0.50000	11	+0.38547	11	+0.38547	3	-0.09432
16	16	+0.50000	11	+0.36486	11	+0.36486	3	-0.10376
17	17	+0.50000	12	+0.37706	12	+0.37706	4	-0.01070
18	18	+0.50000	12	+0.35972	12	+0.35972	4	-0.02038
19	19	+0.50000	13	+0.37112	13	+0.37112	5	+0.04660
20	20	+0.50000	13	+0.35620	13	+0.35620	5	+0.03724
21	21	+0.50000	14	+0.36682	14	+0.36682	6	+0.08852
22	22	+0.50000	15	+0.37610	15	+0.37610	5	+0.02068
23	23	+0.50000	16	+0.38426	16	+0.38426	6	+0.07144
24	24	+0.50000	16	+0.37242	16	+0.37242	6	+0.06379
25	25	+0.50000	16	+0.36135	16	+0.36135	6	+0.05664

28.3 RHC survivor counts

N	3^N	ternary RHC	squarefree RHC	multiplicative RHC	$+\mu(2)$ RHC
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	548	7	3	0
8	6,561	1,854	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	34,938	37	7	0
12	531,441	115,440	37	7	0
13	1,594,323	376,196	46	10	0
14	4,782,969	1,212,672	176	16	0
15	14,348,907	3,874,980	232	10	0
16	43,046,721	7,253,118	232	10	0
17	129,140,163	23,469,114	299	15	0
18	387,420,489	75,346,590	299	15	0
19	1,162,261,467	240,301,818	1,093	53	8
20	3,486,784,401	762,080,202	1,093	53	8
21	10,460,353,203	2,405,119,464	1,471	33	4
22	31,381,059,609	7,558,657,992	4,944	56	8
23	94,143,178,827	23,667,691,934	6,885	82	8
24	282,429,536,481	73,869,142,280	6,885	82	8
25	847,288,609,443	150,937,165,552	6,885	82	8

28.4 RHFc$_{0.05}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	548	7	3	0
8	6,561	846	7	3	0
9	19,683	3,014	7	3	0

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
10	59,049	10,386	8	2	0
11	177,147	34,938	37	7	0
12	531,441	115,440	37	7	0
13	1,594,323	200,592	46	10	0
14	4,782,969	672,132	56	8	0
15	14,348,907	2,220,600	232	10	0
16	43,046,721	7,253,118	232	10	0
17	129,140,163	23,469,114	299	15	0
18	387,420,489	75,346,590	299	15	0
19	1,162,261,467	141,568,470	378	23	0
20	3,486,784,401	459,240,570	378	23	0
21	10,460,353,203	1,479,280,662	1,471	33	4
22	31,381,059,609	4,736,045,276	1,941	26	0
23	94,143,178,827	15,082,716,810	6,885	82	8
24	282,429,536,481	47,810,887,112	6,885	82	8
25	847,288,609,443	150,937,165,552	6,885	82	8

28.5 RHFc$_{0.10}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	226	7	3	0
8	6,561	846	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	16,854	9	3	0
12	531,441	58,812	9	3	0
13	1,594,323	200,592	46	10	0
14	4,782,969	672,132	56	8	0
15	14,348,907	1,157,538	67	5	0
16	43,046,721	3,922,206	67	5	0
17	129,140,163	13,105,098	299	15	0
18	387,420,489	43,282,686	299	15	0
19	1,162,261,467	141,568,470	378	23	0
20	3,486,784,401	257,959,890	378	23	0
21	10,460,353,203	851,645,150	470	17	0
22	31,381,059,609	2,788,013,542	1,941	26	0
23	94,143,178,827	9,060,478,368	2,517	40	0
24	282,429,536,481	29,257,136,160	2,517	40	0
25	847,288,609,443	93,945,019,352	2,517	40	0

28.6 RHFc$_{0.20}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	2	1	1	0
4	81	10	1	1	0
5	243	12	1	1	0
6	729	56	1	1	0

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
7	2,187	226	7	3	0
8	6,561	314	1	1	0
9	19,683	1,250	1	1	0
10	59,049	1,782	1	1	0
11	177,147	7,020	9	3	0
12	531,441	26,340	9	3	0
13	1,594,323	39,860	10	4	0
14	4,782,969	149,610	11	3	0
15	14,348,907	544,068	67	5	0
16	43,046,721	1,929,774	67	5	0
17	129,140,163	3,125,622	79	8	0
18	387,420,489	11,148,402	79	8	0
19	1,162,261,467	39,021,138	92	12	0
20	3,486,784,401	64,683,150	92	12	0
21	10,460,353,203	227,686,670	106	11	0
22	31,381,059,609	789,507,890	576	14	0
23	94,143,178,827	2,702,576,730	697	22	0
24	282,429,536,481	4,644,553,260	697	22	0
25	847,288,609,443	15,993,177,852	697	22	0

28.7 Immediate finite observations

The benchmark separates three effects that should not be conflated. First, the number of admissible completions can collapse by many orders of magnitude while the extremal envelope does not move. In particular, at $N = 25$ the squarefree-support layer has 32768 words and the multiplicative layer only 512, yet both have $B_{25} = 16$. Second, the Möbius-specific orientation $\mu(2) = -1$ changes the extremal envelope sharply, from 16 to 6, leaving eight endpoint RHC completions but no $\mathrm{RHFc}_{0.10}$ completion. Third, these are endpoint statements only; no finite survivor is by itself an RH counterexample. The purpose of the benchmark is to measure deterministic contraction of the completion space and of its adversarial envelope.

29 Conclusion

The surrounding literature already contains most of the basic ingredients: adversarial discrepancy, partial multiplicativity, Chudakov-type rigidity and classification, finite modifications of character-like prime data, structural inverse theorems for multiplicative functions with small sums, random multiplicative fluctuation theory, Möbius sign-pattern results, and conditional information about $M(x)$. The viable research question is therefore narrower. It asks whether a hierarchy of explicitly closed, Möbius-specific partial-information sets has a quantitatively contracting adversarial envelope, and whether the contraction can ever force all fixed-power RHF candidates to disappear without simply revealing the Möbius sequence itself. This document is intended to remain the literature and specification layer for that question.

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